

Infant Elfrick Gnanasusairaj

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EDUCATION

University of Wisconsin (4.0/4.0)

Bachelor of Aerospace Engineering + Computer Science

Madison, WI

Sept. 2025 – Present

EXPERIENCE

Design Innovation Lab AED Drone Dev

Nov 2025 – Present

UW - Madison

Madison, WI

- Worked in a research lab to improve emergency response during ski races, motivated by a recent cardiac fatality at the American Birkebeiner. Led the software development for a drone-based rescue system, in partnership with DJI, enabling rapid and reliable delivery of medical aid by overcoming the access and response-time limitations of snowmobile-based rescue.
- To make AED drone deployment accessible to non-expert users, designed a simple, user-friendly mobile application integrated with DJI's Mobile SDK (MSDK). Implemented a robust backend path-planning system that supports user-defined no-fly zones and computes optimal flight paths using the A* algorithm.
- Collaborated with the lab director on a technical paper describing the project's methodology, system design, and performance results. Contributed analysis, discussion of impact, and proposed directions for future work.

FTC Team Captain

Sept 2018 – Sept 2025

Techno Maniacs

Acton, MA

- Founded and led an FTC robotics team to establish the community's first competitive robotics program. Grew membership from 4 to 20+ students by securing full funding, mentoring new members, and building a culture of continuous iteration. Built a knowledge base from scratch, leading the team from last-place finishes to top 50 globally (of 7,000+ teams) with consistent top-10 New England results.
- To address early under performance and a nonexistent knowledge base, independently built technical depth through rapid iteration, trial-and-error, and targeted outreach to industry experts. Applied this learning across advanced robotics hardware and software, including CAD/CAM, digital fabrication, localization, state machines, and computer vision.
- Built partnerships with companies, schools, and libraries to support the team's growth and outreach. Managed fundraising and outreach early on, then transitioned ownership to an newly-formed business team through structured delegation. These efforts enabled team funding and events that introduced robotics to over 5,000 students.

PROJECTS

WiscHumanoids | Onshape, FEA, ROS, Isaac Sim

Jan 2026 – Present

- Managed the ankle development subteam within a humanoid robotics club. Facilitated research, discussion, and consensus-building around industry standards, ensuring the team worked from a common technical foundation.
- Applied advanced simulation tools to improve both mechanical and software performance. Used FEA to optimize structural designs and NVIDIA Isaac Sim to enable software testing, reducing iteration time and improving system reliability.

Open Source Intake | Java, Android, State Machines, Onshape, 3D Printing, CAM, CNC

Dec 2024 – May 2025

- Designed and released a modular, open-source active intake system for the FTC Into the Deep game to lower the barrier to competitive robot design. Optimized roller geometry and flexible mounting enabled fast, reliable pickup without precise alignment across diverse drivetrains.
- Shared full CAD, build documentation, and usage guidelines to give back to the FTC community. Teams adapted and integrated components into their own systems, providing feedback that drove iterative updates and adoption by 100+ teams worldwide, including 3+ at the 2025 World Championship.

SKILLS [STRONG] [STANDARD]

Languages: [Java, Python, HTML/CSS, JavaScript] [Kotlin, C++, C#, TypeScript, Bash]

Concepts: [OOP, Localization, Path Planning, PID] [Sensor Fusion, State Machines, Kinematics]

Frameworks: [React, Android, Unity, Node.js, Next.js] [ROS, Linux, Material-UI, Tailwind, Arduino]

Tools: [Git, Visual Studio, JetBrains IDEs, Onshape, 3D Printing, CNC, Fusion 360] [GCP, Blender, FEA, Isaac Sim]

Libraries: [TensorFlow, OpenCV, DJI MSDK, NumPy] [pandas, PyTorch, face detection, Matplotlib, NLTK]